



LarvaLens

Technical Requirements Document - TRD

Locked stack, model-loading rules, environment variables, frontend/backend constraints, security controls, performance budget and technical completion criteria.

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Vibe Document 02 - Technical Requirements Document

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7.1 Technology decisions

Concern	Locked choice
Web	React, Vite, TypeScript, React Router
UI	Tailwind CSS, accessible primitives, Lucide icons
Client data	TanStack Query; Supabase JS only for auth, authorized DB reads and Realtime
PWA	Web app manifest + service worker; cache shell assets, never cache private videos/results publicly
API	Python 3.11+, FastAPI, Pydantic v2, Uvicorn
ML/CV	Ultralytics, PyTorch, OpenCV, NumPy
Database	Supabase PostgreSQL
Auth	Supabase Auth email/password; JWT forwarded to FastAPI
Storage	Private <code>scan-videos</code> and <code>scan-evidence</code> buckets
Realtime	Supabase Postgres Changes on the signed-in user's scan row
Map	Leaflet + OpenStreetMap tiles with attribution
Frontend deployment	Vercel
Inference deployment	Long-running local/LAN or Python service; no assumption of GPU on Vercel
Tests	Pytest + FastAPI TestClient; Vitest + React Testing Library; Playwright if time permits

7.2 Model-loading rules

- Backend reads `MODEL_DIR/model_manifest.json` during application lifespan.
- It validates the two required files and SHA-256 values before setting `models_ready=true`.
- Models load once per process, not per request.
- A model reload is an explicit administrator/local operation; no arbitrary file path comes from a request.
- A scan stores model hashes copied from the loaded manifest.
- Missing, corrupt or mismatched artifacts produce `MODEL_NOT_READY`; never fall back to a remote AI API.

7.3 Environment variables

text

```
# Web - public values only
VITE_API_BASE_URL=
VITE_SUPABASE_URL=
VITE_SUPABASE_ANON_KEY=

# API - secrets stay server-side
APP_ENV=development
APP_ORIGIN=http://localhost:5173
SUPABASE_URL=
SUPABASE_ANON_KEY=
SUPABASE_SERVICE_ROLE_KEY=
MODEL_DIR=/absolute/path/to/models/active
MAX_VIDEO_MB=80
MAX_VIDEO_SECONDS=15
```

```
ALLOWED_VIDEO_MIME_TYPES=video/mp4,video/webm,video/quicktime
ENABLE_SPECIES_MODEL=false
LOG_LEVEL=INFO
```

Never prefix `SUPABASE_SERVICE_ROLE_KEY` with `VITE_`. Never commit `.env`, Kaggle tokens, private videos or `.pt` weights.

7.4 API engineering constraints

- Version routes under `/api/v1`.
- Use Pydantic response models; no untyped arbitrary dictionaries at the route boundary.
- Validate MIME type, extension, file signature, size and decodable duration.
- Generate storage paths server-side from authenticated user/scan UUIDs.
- Use idempotency key support for upload retries.
- Return structured errors: `code`, `message`, `retryable`, `request_id`.
- Heavy inference runs outside the request handler. Return HTTP 202 after the real scan row and upload succeed.
- Use bounded concurrency so two large videos do not exhaust memory.
- Delete temporary local copies in `finally` after storage/evidence persistence.
- Log IDs and timings, not tokens or private video contents.

7.5 Frontend engineering constraints

- Route guards wait for auth hydration.
- Every async page has loading, empty, error and success states.
- Upload UI shows real client-side validation before sending.
- Progress UI subscribes to the user's authorized scan row and also refetches on reconnect.
- Do not display an analysis card until the corresponding real row exists.
- Map handles denied geolocation and scans without coordinates.
- Keyboard navigation, visible focus, labels, contrast and reduced-motion support are required.

7.6 Security requirements

- RLS enabled on every public table.
- Authorization roles live in `profiles.role` or trusted app metadata, not editable user metadata.
- Service role used only in API process.
- Private buckets; signed URLs expire quickly and are issued only after row authorization.
- Public hotspot endpoint returns rounded/aggregated coordinates and counts, not raw scans.
- Raw model evidence is immutable to ordinary users; reviewer decisions are separate records.

7.7 Performance budget

Area	MVP budget
Web initial load	Keep core route lean; lazy-load map/reviewer pages.
Upload	Up to configured 80 MB and 15 seconds; reject before inference if exceeded.
Model load	Once during API startup. Health exposes readiness, not file paths.
Processing	Sample frames as configured; maintain bounded in-memory crops.
Realtime	One filtered subscription for the active scan; unsubscribe on route exit.

7.8 Definition of technical completion

The system is technically complete only when an authenticated user uploads a real video, the API stores it privately, the supplied model hashes pass, inference runs, the database receives real track evidence, the PWA updates from real status changes, and the final screen/map use those rows.